DIPLOMA IN SOFTWARE ENGINEERING | DEGREE YEAR-1-21.1

STUDENT HAND BOOK

NATIONAL INSTITUTE OF BUSNINESS MANAGEMENT

NATIONAL INSTITUTE OF BUSINESS MANAGEMENT

NIBM is statutory body functioning under the Ministry of Higher Education, Technology and Innovation. With over 52 Years of experience in the Sri Lanka IT education field NIBM provides services in the areas of management education, IT training and consultancy.

The entire operations of the institute are monitored by the Governing Council and all educational offerings are centered under two divisions, namely, Productivity and Management Development Division and Management Information Systems Division.

NIBM Undoubtedly has given an impetus to the nation through its progressive march towards development. The institute is well equipped with modern teaching aids and a conductive learning environment for student to acquire knowledge without any disturbance.

SCHOOL OF COMPUTING AND ENGINEERING

The School of Computing and Engineering conducts academic programs, professional certifications programs, computer training programs and engineering programs, which consists of several Degree Programs, Diploma Programs, a Higher National Diploma Programs and several Certificate levels programs.

These Programs have gained wide acceptance by the public as a result of begin well-structured to cater to the needs of the participants. Due to its long experience in the field, the NIBM is recognized as the pioneer computer training institute in the country.

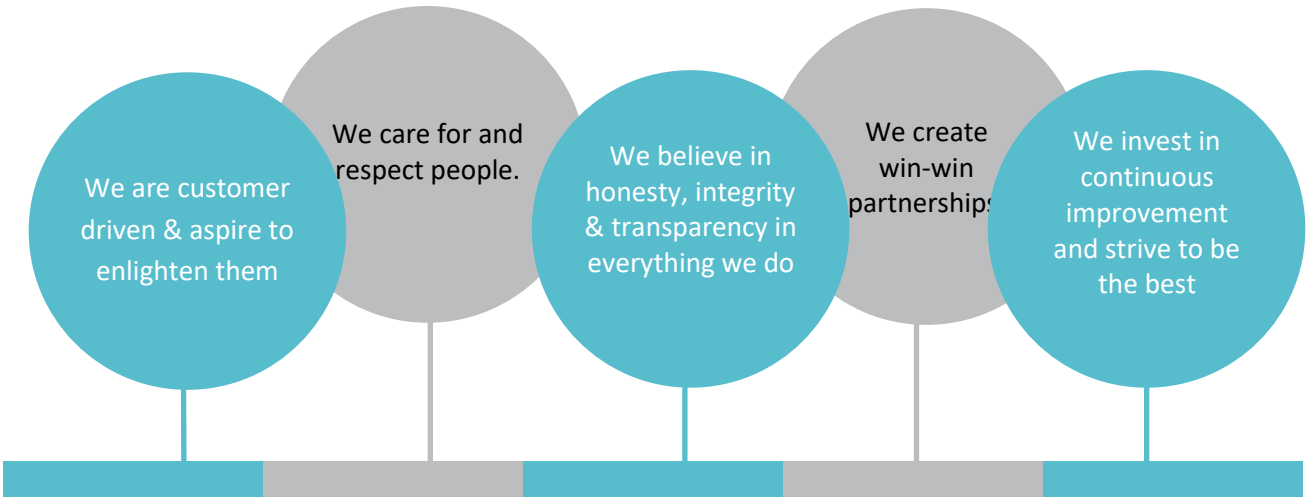
OUR VISION

"To be the best business education institute in Sri Lanka"

OUR MISSION

"We are in the business of developing competencies in people & organizations through training and consultancy"

OUR VALUES



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DISCIPLINARY PROCEDURE

NIBM values following disciplines of the students during the program. The participants who violate NIBM disciplinary procedure will be taken a disciplinary action by NIBM.

- Punctuality**.**
  - The participant should arrive and leave for lectures, course works (Class Tests) and written examinations on time. [Refer relevant schedules]
  - The participant should submit course works of modules on time.
- Behavior.
  - The participant should behave in a well manner within lecture hall and NIBM premises.
  - The conflicts are not expected between students and between NIBM staff and students.
- Administration.
  - Program Fee.

    The participant should pay program fee according to given date(s) in the letter offered by program office. If not NIBM will drop the participant immediately.
  - Study Materials.

    The participants are not distributed printed study materials by NIBM. The study martials are uploaded on NIBM WORLD WIDE WEB. (LMS)
  - Written Examination

    The students are adhered to the NIBM examination procedure mentioned in **Appendix-1**.
  - Computer Laboratory

    The participants should follow lab procedure mentioned in **Appendix-2**.
- Immediate contact.

The participant should inform to program officer/course director/director in emergency.

PROGRAM SPECIFICATION

OVERVIEW AND PROGRAMME AIMS

The Diploma in Software Engineering is a highly recognized Diploma program offered by the National Institute of Business Management (NIBM). This diploma is newly added to the curriculum as the industry demand is highly increasing in the field of Software Engineering. It is widely recognized all over the world as an academic qualification which fulfills entry requirements to a wide range of degree earning pathways as well as a prime professional qualification which would provide superior recognition for practicing computer professionals. The program is conducted for one academic year on full time or Part time basis, covering more than 600 hours of classroom teaching as well as practical sessions and a Software Development Project.

The Diploma in Software Engineering (Full Time) Program is especially designed for school leavers and the Part time program is designed for working professionals seeking an academic qualification.

The Aims of the Diploma in Software Engineering are:[1]

- To provide knowledge and skills in computer programming and software Engineering required to develop Information Systems used in Business and Industrial environments
- To prepare participants to start a career in the IT industry
- To provide a basic understanding of the current position and challenges of the IT industry
- To satisfy the academic criteria required for admission to degree programs in IT and to obtain membership of professional institutions.

PROGRAM LEARNING OUTCOMES

On successful completion of this program the participant will be able to:

→ Engineer Requirements

→ Design software

→ Program software

→ Assure Quality

→ Maintain Software

PROGRAM STRUCTURE

The DSE is a program with total credit weighting amounting to 45. A credit is defined as the workload of 15 contact hours.

A minimum of 100 hours of study is recommended for each module. Also, a minimum 200 hours of study recommended for project

MODULES AND CREDITS

Module Code	Module Name	Credit Value
Credit Based Modules		
	Semester-1	
NIBM-SOC-COM -1-1-3-01	Introduction to Computer Science	3
NIBM-SOC-COM-1-1-3-02	Mathematics for Computing	3
NIBM-SOC-COM-1-1-3-03	Programming Fundamentals	3
NIBM-SOC-COM-1-1-3-04	Electronics and Computer Architecture	4
NIBM-SOC-COM-1-1-3-05	Object Oriented Programming	2
NIBM-SOC-COM-1-1-3-06	Database Management Systems	4
	Semester-2	
NIBM-SOC-COM-1-2-4-07	Computer Networks	3
NIBM-SOC-COM-1-2-3-08	GUI Application Development	4
NIBM-SOC-COM-1-2-4-09	Software Engineering	4
NIBM-SOC-COM-1-2-3-10	Enterprise Application Development-1	4
NIBM-SOC-COM-1-2-4-11	Operating Systems	4
NIBM-SOC-COM-1-2-4-12	Developing Modern Web	4
NIBM-SOC-COM-1-2-5-13	Business Solution Development Project	3
ZERO-CREDIT BASED MODULES		
NIBM-SOC-COM-B-0-14	Effective Communication Skills-1	0
	Total Credits	**45**

TEACHING AND LEARNING METHODS

The program contents are delivered through lectures, classroom discussions, presentations, practical sessions and group projects. The material for all modules except independent project is taught through lectures. In most cases supporting materials, including notes, slides and sometimes videos, are made available to participants. Lecturers also direct participants to recommended reading to supplement the lecture material.

EVALUATION METHODS

The module evaluation methods are **Written Examination** and **Course Work**. The evaluation criteria for each module is given in the module evaluation criteria section under each module.

The participant should obtain **40%** for each module evaluation method [Written Examination and Course work], to complete the module successfully.

If a participant fails a module method(S) [Written Examination or Course work], The participant should re-sit and complete the module evaluation method(s) with same batch re-sit schedule(s).

If the participant is unable to complete module method(s) with same batch, he or she must repeat all module methods with an upcoming batch.

Participants are expected to maintain at least **70%** attendance for each module to be eligible to sit for Written Examination.

NIBM shall have the final deciding authority in this purpose.

GRADING CRITERIA

A candidate's performance in each module will be graded according to the following grading scheme.

Range of Marks	Grade	Grade Point
85-100	A+	4.0
70-84	A	4.0
65-69	A-	3.7
60-64	B+	3.3
55-59	B	3.0
50-54	B-	2.7
45-49	C+	2.3
40-44	C	2.0
35-39	C-	1.7
30-34	D+	1.3
25-29	D	1.0
00-24	E	0.0

GRADE POINT AVERAGE

$$GPA = \frac{\sum Gi.Ci}{\sum Ci}$$

Grade Point Average (GPA) is the credit-weighted arithmetic mean of the Grade Point Values, which is determined by dividing the total credit-weighted Grade Point Value by the total number of credits. GPA shall be computed to the ***second decimal place***.

Where,

G_i *=Grade Point* C_i*=Number of Credits*

DIPLOMA AWARDING CRITERIA

Eligibility for the award of Diploma in Computer System Design are:

a) Obtain **C** grade for **each module and**
b) Achieve an overall **GPA of 3.3** or better **and**
c) Pass Effective Communication Skills-1 Module Final Semester Evaluation (**50/100**)

DISTINCTION

Candidates who obtain an overall **GPA equal or above 3.80 in the first attempt** will be awarded the Diploma in Computer System Design with a Distinction.

GPA	Grade
Equal or above 3.30 up to 3.79	Pass
Equal or above 3.8	Distinction

MODULE DESCRIPTOR

INDUCTION

Induction session helps student to understand the Current Technologies & Future of ICT Industry, how to maintain the Professional Ethics & Attitudes, Report Writing Techniques, Effective Presentation Techniques. Other than that student learns the history of the Computer and current ICT Laws. The knowledge gained in this module is helpful for participants to get an overall knowledge about the Current Technologies and Future of the Industry.

LEARNING OUTCOMES

At the end of this module, students should be able to:

1. Describe DCSD Course Overview
2. Explain Platform Revolution
3. Explain Communication Ethics
4. Apply Report Writing techniques
5. Describe Current Technologies and Future IT Industry
6. Explain Lifestyle During Pandemic Situations
7. Apply Effective Presentation Techniques
8. Apply Fundamentals for Assignments
9. Describe History of the Computer
10. Explain ICT Laws

LECTURE SESSIONS

No	Session	Teaching Hours	Learning Outcomes	Method
1	DCSD Course Overview	2	1	Lecture
2	Platform Revolution	2	2	Lecture
3	Communication Ethics	2	3	Lecture
4	Report Writing techniques	3	4	Lecture|Lab
5	Current Technologies and Future IT Industry	3	5	Lecture
6	Lifestyle During Pandemic Situations	3	6	Lecture
7	Effective Presentation Techniques	3	7	Lecture|Lab
8	Basic Fundamentals for Assignments	3	8	Lecture|Lab
9	History of the Computer	3	9	Lecture
10	ICT Laws	3	10	Lecture
Total		**27**		

INTRODUCTION TO COMPUTER SCIENCE

INTRODUCTION

The computer science is a highly demanded and growing field in the world today. This module covers the fundamentals of computer hardware and software and advanced concepts such as security, networking, multimedia, program logic modeling and basics of computer architecture and the responsibilities of an IT professional. The knowledge gained in this module is helpful for participants to study key areas of computer science in depth.

LEARNING OUTCOMES

At the end of this module, students should be able to:

1. Explain the different number systems and how data is represented within the computer system
2. Select the appropriate computer components to build, repair, or upgrade personal computers
3. Explain how to correctly use tools and safely work in a lab
4. Explain Software and how it executes on software
5. Explain the concept of network communications
6. Configure devices to connect to the Internet
7. Explain basic concepts and use of Multimedia
8. Apply the logic for computer programs
9. Describe the basics of computer architecture

LECTURE SESSIONS

No	Session	Teaching Hours	Learning Outcomes	Method
1.1	Number Systems and Data Representation	12	1	Lecture|Tute
1.2	Computer Hardware	6	2,3	Lecture|Lab
1.3	Computer Software	3	4	Lecture|Lab
1.4	Computer Network and Internet	6	5,6	Lecture|Lab
1.5	Multimedia Concepts	6	7	Lecture|Lab
1.6	Flow Charts & Pseudo Codes	12	8	Lecture
Total		**45**		

PRACTICAL SESSIONS

No	Session	Practical Hours
1	PC Assembly	3
2	Working with operating systems and software	6
3	Working with the network	6
4	Security best practices	3
Total		**18**

MATHEMATICS FOR COMPUTING

INTRODUCTION

Mathematics are applied in various computing applications. This module covers the basic mathematical concepts which are required in a computing degree. After successful completion of the module, participants should be able to apply mathematical concepts to solve problems in the areas of vectors, matrices, statistics, elementary probability and statistics.

LEARNING OUTCOMES

At the end of this module, students should be able to:

1. Solve simultaneous and quadratic equations.
2. Familiarize with matrices including matrix multiplication, row operations, determinants and applications including the solution of linear equations.
3. Formulate problems in the language of sets and perform set operations.
4. Differentiate between combinations and permutation along with specific usage.
5. Compute probabilities based on the practical situations using probability distributions.
6. Define events and assign probabilities to them.
7. Define Measurers of statistical variability.

LECTURE SESSIONS

No	Session	Teaching Hours	Learning Outcomes	Method
2.1	Basic Algebra	06	1	Lecture|Tute
2.1	Matrices	09	2	Lecture|Tute
2.2	Sets and Relations	06	3	Lecture|Tute
2.3	Combinations and Permutations	06	4	Lecture|Tute
2.5	Probability	09	5,6	Lecture|Tute
2.4	Descriptive Statistics	09	7	Lecture|Tute
Total		**45**		

MODULE EVALUATION CRITERIA

Method	Percentage	Learning outcome
CW-1 [Class Test]	30%	1,2,3,4
Written Exam [3 Hours]	70%	1,2,3,4,5,6,7

RECOMMENDED REFERENCES

Book	Author(s)	ISBN	Publisher
Quantitative Methods for Computer Students	B. Catlo	1873981171	Wolfe Publishing
Business Statistics	Ken Black	0324009208	South-West College

PROGRAMMING FUNDAMENTALS

INTRODUCTION

The programming is a crucial task in software engineering which is used to write software instructions. This module is designed to introduce fundamentals of software programming for the participants. After study of this module, participants are capable to design program logics using problem solving techniques and write software instructions applying programming fundamentals.

LEARNING OUTCOMES

At the end of this module, students will be able to:

1. Explain computer programming and belongings.
2. Use data types with variables.
3. Apply operators to process data.
4. Apply selection techniques to make programing decisions.
5. Apply repetition techniques for re-executions.
6. Apply arrays in programming to handle multiple data elements.
7. Use of pointers and manage memory.
8. Apply functions in programming for handling several segments.
9. Use of files for storing and retrieving data.
10. Apply best practices for programming software.

LECTURE SESSIONS

No	Session	Teaching Hours	Learning Outcomes	Method
3.1	Introduction to Software Programming	3	1	Lecture|Lab
3.2	Data Types and Variables	3	2	Lecture|Lab
3.3	Data Operators	3	3	Lecture|Lab
3.4	Selections	6	4	Lecture|Lab
3.5	Repetitions	6	5	Lecture|Lab
3.6	Arrays	6	6	Lecture|Lab
3.7	Pointers and Memory	3	7	Lecture|Lab
3.8	Functions	6	8	Lecture|Lab
3.9	File Operations	6	9	Lecture|Lab
3.10	Best Coding Practices	3	10	Lecture|Lab
Total		**45**		

MODULE EVALUATION CRITERIA

Method	Percentage	Learning Outcomes
Course Work-1[Practical Test] [I]	50%	1,2,3,4,5,6,7,8,9,10
Written Exam [2 Hours]	50%	1,2,3,4,5,6,7,8,9,10

RECOMMENDED REFERENCES

Book	Author(s)	ISBN	Publisher
The C Programming Language	Brian W.Kernighan & Dennis M.Ritchie	8120305965	Prentice Hall
C How to Program	H.M.Deitel & P.J.Deitel	0132883333	Prentice Hall

ELECTRONICS AND COMPUTER ARCHITECTURE

INTRODUCTION

This module contains two main areas which are electronics and computer architecture. The electronics part covers Fundamental physical electronic concepts and digital electronic concepts which are required for computer science. The computer architecture part covers microprocessor architecture , memory organization and input output interfacing techniques used in a computer system.

LEARNING OUTCOMES

At the end of this module, students should be able to:

1. Apply rules of Boolean Algebra, Karnaugh map method and Tabular method to simplify Boolean expressions.
2. Explain the basic concepts of digital electronics.
3. Design basic digital electronic circuits for practical applications.
4. Explain analog and digital signal conversions.
5. Explain relationship between digital and analog electronic components
6. List down different subsystems and their components in a computer and explain their functionalities.
7. Draw a sketch of microprocessor and identify its components.
8. Write Assembly and Machine programs and understand the execution of instructions in processor architecture.
9. Explain function of the control unit and it's implementation
10. Understand memory organization and data storage techniques in computer system.
11. Describe IO interfacing in a computer system and discuss IO usage in embedded systems.

LECTURE SESSIONS

No	Session	Teaching Hours	Learning Outcomes	Method
4.1	Fundamentals of Digital Circuit Design	6	1	Lecture|Lab
4.2	Logic Gates and Design of Combinational Logic Circuits and Usage of Logic ICs.	3	2,3	Lecture|Lab
4.3	Multiplexers, Demultiplexers and Shift Registers.	3	2	Lecture|Lab
4.4	Sequential Circuit Design.	6	3	Lecture|Lab
4.5	Signal Conversions.	3	4	Lecture| Lab
4.6	RTL, DTL, TTL, ECL, MOS technologies and comparison of properties of Logic Families.	9	5	Lecture| Lab
4.7	Computer Organization and Microprocessor Architecture	6	6,7	Lecture|Tute
4.8	Assembly and Machine Programming	6	8	Lecture|Tute
4.9	Control unit implementation	6	9	Lecture|Tute
4.10	Computer Memory Techniques	6	10	Lecture|Tute
4.11	Input Output interfaces	6	11	Lecture|Tute
	Total	**60**		

MODULE EVALUATION CRITERIA

Method	Percentage	Learning outcome
Course Work-1[App][G]	50%	1,2,3,4,5
Written Exam [2 Hours]	50%	1,2,3,4,5,6,7,8,9,10,11

RECOMMENDED REFERENCES

Book	Author(s)	ISBN	Publisher
Electronic Devices and Circuits	T.F. Bogart	0133937607	Prentice-Hall
Digital Design	M. Morris Mano	8120320956	Prentice-Hall
Modern Digital Electronics	R P Jain	0070494924	Tata McGraw Hill

OBJECT ORIENTED PROGRAMMING

INTRODUCTION

The Software is developed by integrating multiple objects according to the object-oriented concept. This module introduces object-oriented concept and its principles for developing object-oriented software applications. Participants are able to apply object-oriented concepts for programming software applications.

LEARNING OUTCOMES

At the end of this module, students should be able to:

1. Construct programs using JAVA language.
2. Explain the core concepts of Object-Oriented Programming.
3. Apply object-oriented concept with classes and objects.
4. Use of Inheritance in object-oriented programming.
5. Use of Polymorphism in object-oriented programming.

LECTURE SESSIONS

No	Session	Teaching Hours	Learning Outcomes	Method
5.1	Language Features	3	1	Lecture|Lab
5.2	Object-Oriented Concept	3	2	Lecture
5.3	Classes and Objects	6	3	Lecture|Lab
5.4	Association	3	3	Lecture|Lab
5.5	Inheritance	9	4	Lecture|Lab
5.7	Polymorphism	6	5	Lecture|Lab
	Total	**30**		

MODULE EVALUATION CRITERIA

Method	Percentage	Learning Outcomes
Course Work-1[Practical Test]	50%	1,2,3,4,5
Written Exam [2 Hours]	50%	1,2,3,4,5

RECOMMENDED REFERENCES

Book	Author(s)	ISBN	Publisher
Object Oriented Programming with C++	E.Balagurusamy	9780070669079	Tata McGraw Hill
Teach yourself C++	Herbert Schildt	007463870X	Tata McGraw Hill

DATABASE MANAGEMENT SYSTEMS

INTRODUCTION

Database is a key component of most of software for storing data. This module focuses on designing and implementing databases require for software application. The participants learn to design database using diagrammatic way and to convert that into a physical database. Students are able to complete **oracle academy certification** at the end of the module.

LEARNING OUTCOMES

At the end of this module, students should be able to:

1. Describe database system concepts and architecture
2. Apply ER/EER diagrams for a real-world problem scenario
3. Map the ER/EER to the Relational Schema
4. Apply Normalization rules to Normalize relations in a database and File Designing
5. Apply SQL DDL statements with Constraints to develop database and table (Create, Alter, Describe, Drop, Truncate etc.)
6. Apply SQL DML statements to maintain the table (Select, Insert, Update, Delete, Aggregate Functions, Group by, Having etc.)
7. Apply SQL Joining, Views and Sub Queries to display the data from different tables.
8. Apply Stored Procedures, Triggers and Functions in SQL Server.
9. Apply Database Security (DCL) statements by creating different Users, Schemas and Roles.
10. Apply Row Level Security, Encryption, Permissions for different user levels.
11. Explain Database Auditing (Failed logins, SQL Server login changes etc.)
12. Backups and Restore Database

LECTURE SESSIONS

No	Session	Teaching Hours	Learning Outcomes	Method
6.1	DBMS Concepts and Architecture	3	1	Lecture
6.2	Conceptual Design (ER|EER)	6	2	Lecture|Tute
6.3	Schema Mapping	6	3	Lecture|Tute
6.4	Normalization	6	4	Lecture|Tute
6.5	SQL DDL statements with Constraints	6	5	Lecture|Lab
6.6	SQL DML statements to maintain the table	9	6	Lecture|Lab
6.7	SQL Joining, Views and Sub Queries	6	7	Lecture|Lab
6.8	Stored Procedures, Triggers and Functions	3	8	Lecture|Lab
6.9	Database Security (DCL) statements with creating different Users, Schemas and Roles	6	9	Lecture|Lab
6.10	Row Level Security, Encryption, Perimissions for differnet user levels	6	10	Lecture|Lab
6.11	Database Auditing, Backups and Restore Database	3	11, 12	Lecture|Lab
	Total	**60**		

MODULE EVALUATION CRITERIA

Method	Percentage	Learning outcome
Course Work 1 [Practical Test]	40%	5,6,7,8,9,10,11
Written Exam [2 Hours]	60%	1,2,3,4,5,6,7,8,9,10,11

RECOMMENDED REFERENCES

Book	Author(s)	ISBN	Publisher
Fundamentals of Database Systems	Elmasri, Navathe,	81-317-0580-3	Pearson Publishing

COMPUTER NETWORKS

INTRODUCTION

This course introduces the architecture, structure, functions, components, and models of the Internet and other computer networks.
This Module introduced the principles of computer networking, IP addressing, and the fundamentals of Ethernet concepts, media, cloud operations.

LEARNING OUTCOMES

At the end of this module, students should be able to:

1. Explain Network Protocols, Communications, OSI Reference model and TCP/IP model
2. Explain how switches and routers operate in a small to medium-sized business network.
3. Design an IP addressing scheme, calculate, and apply subnet masks and addresses
4. Configure initial settings on a network device and Implement basic network connectivity between devices.
5. Describe Routing concepts.
6. Describe Network address translation (NAT).
7. Describe Network Security concepts.
8. Describe Cloud computing concepts.

LECTURE SESSIONS

No	Session	Teaching Hours	Learning Outcomes	Method
7.1	Network Protocols and Communications	6	1	Lecture|Lab
7.2	Network Access and Ethernet	6	1,2	Lecture|Lab
7.3	Network Layer	3	1,2,5	Lecture|Lab
7.4	IP Addressing and subnetting	6	3	Lecture|Lab
7.5	Transport and Application Layer protocols	6	1	Lecture|Lab
7.6	Routing Concepts	3	5	Lecture|Lab
7.7	NAT	3	6	Lecture|Lab
7.8	Network Security	3	7	Lecture|Lab
7.9	Cloud Concepts	9	8	Lecture|Lab
	Total	**45**		

MODULE EVALUATION CRITERIA

Method	Percentage	Learning outcome
Course Work-1[Class Test]	25%	
Course Work-2[Practical Test]	25%	
Written Exam [2 Hours]	50%	

RECOMMENDED REFERENCES

Book	Author(s)	ISBN	Publisher
Data and Computer Communications(10th Edition)	Stallings , W	1292014385	Pearson

GUI APPLICATION DEVELOPMENT

INTRODUCTION

GUI software applications are well demand. This module introduces technologies which are used to develop Graphical User Interface software applications. The participants are able to develop user-friendly GUI applications using available software tools and technologies.

LEARNING OUTCOMES

At the end of this module, student should be able to:

1. Explain .Net architecture and C# program structure, Syntax and Basic Object-oriented concepts using Console
2. Develop Windows Forms using Object Oriented Concepts, Classes and Business Logics
3. Develop Windows Forms with different NuGet Packages
4. Develop Windows Presentation Foundation (WPF) with XAML
5. Apply Input Validation and Exception Handling with Windows Presentation Foundation (WPF)
6. Connect WPF/Windows Forms and Database using ADO.NET with Classes and Objects
7. Connect WPF/Windows Forms and Database using Language Integrated Query (LINQ) and Object Relational Mapping with Entity Framework
8. Programs with Cloud platforms (Azure/Amazon)
9. Apply Microsoft/Crystal Reports and Setup creation
10. Programs with SOAP and RESTful web services and API
11. Introduction to .NET Core architecture and .NET Standard Libraries

LECTURE SESSIONS

No	Session	Teaching Hours	Learning Outcomes	Method
8.1	C# program structure, Syntax using Console	3	1	Lecture
8.2	Basic OOP Concepts using Console Applications	6	1	Lecture|Lab
8.3	Windows Forms with Basic Controls, Classes and Business Logics	9	2	Lecture|Lab
8.4	Windows Forms with NuGet Packages	3	3	
8.5	Windows Presentation Foundation with Exception Handling and Validations	9	4,5	Lecture|Lab
8.6	ADO.NET with Classes and Objects	6	6	Lecture|Lab
8.7	Language Integrated Query (LINQ) and ORM with Entity Framework	9	7	Lecture|Lab
8.8	Introduction to Cloud computing and Cloud platforms with Azure/Amazon and Microsoft/Crystal Reports	3	8,9	Lecture|Lab
8.9	Introduction to .NET Core architecture	6	10	
8.10	Programming with SOAP and RESTful web services and API	6	11	Lecture|Lab
	Total	**60**		

MODULE EVALUATION CRITERIA

Method	Percentage	Learning outcome
Course Work 1[Software App] [G]	50%	2,3,4,5,6,7,8,9,10,11
Written Exam [2 Hours]	50%	1,2,3,4,5,6,7,10,11

RECOMMENDED REFERENCES

Book	Author(s)	ISBN	Publisher
Head first C#	Jennifer Greene, Andrew Stellman	1449343503	O'Reilly Media
Microsoft Visual C# Step by Step	John Sharp	1509301046	Microsoft Publications

SOFTWARE ENGINEERING

INTRODUCTION

Software Engineering goal is to identify and evaluate importance of software engineering (best practices) to build good software. It aims to develop a broad understanding of software processes, software requirements, important of software design, agile software development, Web Engineering, importance of software maintenance and software configuration. After successfully completion of the module, student should be able to apply software engineering concepts to solve problems in the areas of Information systems and to build good software in business organizations.

LEARNING OUTCOMES

At the end of this module, students should be able to:

1. Apply out of the box thinking into the application design.
2. Describe importance of Software Engineering, best practices and Identify different Software Development Approaches.
3. State different Software Requirements and describe the Software Requirement process.
4. Identify the importance of Agile, Scrum Practices and apply Agile Development for building software.
5. Draw UML for software analysis.
6. Describe importance of Software Testing and its different approaches
7. Use the techniques for controlling different versions of the application.
8. Prepare schedule, cost and budget for the software project and analyze the risk factors.
9. Identify the importance of Software Maintenance and Configuration Management.

LECTURE SESSIONS

No	Session	Teaching Hours	Learning Outcomes	Method
9.1	Conceptual Thinking	3	1	
9.2	Software Engineering and SDLC	3	2	Lecture
9.3	Software Development Approaches	3	2	Lecture
9.4	Software Requirement Engineering	6	3	Lecture
9.5	Agile Techniques	12	4	Lecture
9.6	Software Analysis with UML	9	5	Lecture
9.7	Test Driven Developments and Behavior Driven Developments	6	6	Lecture
9.8	Version Controlling	3	7	Lecture
9.9	Project Management	12	8	
9.10	Software Maintenance	3	9	Lecture
	Total	**60**		

MODULE EVALUATION CRITERIA

Method	Percentage	Learning outcomes
Course Work-1[In Class Test] [G]	50%	1, 2
Written Exam [2 Hours]	50%	1,2,3,4,5,6,7

RECOMMENDED REFERENCES

Book	Author(s)	ISBN	Publisher
Software Engineering A practitioner's approach	Roger Pressman	0078022126	McGraw-Hill Education
Software Engineering	Ian Sommerville	13: 978-0-13-703515-1	Pearson

ENTERPRISE APPLICATION DEVELOPMENT-1

INTRODUCTION

Enterprise applications are used in variety of businesses and organizations. This module is designed to provide knowledge on advanced concepts which are used to develop enterprise level applications. This module benefits participant to apply latest and advanced technologies for developing modern business applications used by enterprises. The students are able to complete oracle academy certification at the end of the module.

LEARNING OUTCOMES

At the end of this module, students should be able to:

1. Apply best styles of UI design for better user experience.
2. Apply advanced UI components for complex applications.
3. Use main programming concepts for software development.
4. Perform database operations.
5. Perform data manipulations with generics and arrays.
6. Use lambda expression for functional programming.
7. Apply design patterns for software developments.
8. Produce reports required for decision making.
9. Generate deployment version of the enterprise application.

LECTURE SESSIONS

No	Session	Teaching Hours	Learning Outcomes	Method
10.1	Best Practices of UI and UX	9	1	Lecture | Lab
10.2	Advanced Swing Components	6	2	Lecture | Lab
10.3	Language Core Concepts	9	3	Lecture | Lab
10.4	Data Driven Application	9	4	Lecture | Lab
10.5	Generics and working with Arrays	6	5	Lecture | Lab
10.6	Lambda Expressions	3	6	Lecture | Lab
10.7	Design Patterns	6	7	Lecture | Lab
10.8	Reports	6	8	Lecture | Lab
10.9	Application Deployment	6	9	Lecture | Lab
	Total	**60**		

MODULE EVALUATION CRITERIA

Method	Percentage	Learning outcome
Couse Work-1[Enterprise App] [I]	50%	1,2,3,4,5,7,8,9
Final Exam [2 Hours]	50%	1,2,3,4,5,6,7,8,9

RECOMMENDED REFERENCES

Book	Author(s)	ISBN	Publisher
Head First Java	Kathy Sierra	9788173666025	O'Reilly Publishing
Java in 60 Minutes a Day Bible	R. F. Raposa	8126504277	Wily Dreamtech India

OPERATING SYSTEMS

INTRODUCTION

Operating systems are an essential part of any computer system. Similarly, a course on operating systems is an essential part of any computer science education. This field is undergoing rapid change, as computers are now prevalent in virtually every arena of day-to-day life from embedded devices in automobiles through the most sophisticated planning tools for governments and multinational firms.

This course provides an understanding of entire process within Operating Systems. It covers overview of Operating Systems, Operating System principles, concurrency, scheduling and dispatch, memory management, security and protection, virtual machines, device management and file systems.

Students are expected to read the assigned materials prior to each class, and to participate in in-class discussions.

LEARNING OUTCOMES

At the end of this module, students should be able to:

1. Identify Operating-System Structure and services provided by an operating system.
2. Identify the separate components of a process and how processes are created and terminated in an operating system
3. Identify the basic components of a thread, and contrast threads and processes.
4. Explain difference between a logical and a physical address and the role of the memory management unit (MMU) in translating addresses.
5. Describe virtual memory benefits and Explain how it works
6. Describe the physical structures of various secondary storage devices
7. Explain the characteristics of mass-storage devices.
8. Explain the principles and complexities of I/O hardware and deadlock handling.
9. Explain the function of file systems in Operating System
10. Describe the virtualization technologies and networked and distributed systems.
11. Discuss Windows, Linux and MAC Operating system and its functionality.
12. Discuss the functionality of mobile operating systems.

LECTURE SESSIONS

No	Session	Teaching Hours	Learning Outcomes	Method
11.1	Introduction	6	1	Lecture
11.2	Processes, Process Synchronization and Threads	6	2,3	Lecture|Lab
11.3	Memory Management	6	4,5	Lecture|Lab
11.4	Storage Management	6	6,7	Lecture|Lab
11.5	Input Output and Deadlocks	6	8	Lecture|Lab
11.6	File System	6	9	Lecture|Lab
11.7	Virtualization and Distributed Systems	6	10	Lecture|Lab
11.8	Windows Linux, MAC Operating System	12	11	Lecture|Lab
11.9	Mobile Operating System	6	12	Lecture|Lab
	Total	**60**		

MODULE EVALUATION CRITERIA

Method	Percentage	Learning outcome
Coursework	50%	All
Exam [2 Hours]	50%	All

RECOMMENDED REFERENCES

Book	Author(s)	ISBN	Publisher
Operating System Concepts(10th Edi)	Abraham Silberschatz, Greg Gagne, Peter B. Galvin	978-1-119-32091-3	Wiley
Modern Operating Systems(4th Edi)	A.S. Tanenbaum	013359162	

DEVELOPING MODERN WEB

INTRODUCTION

Modern web development is a highly demand and growing field that impact the world. This module is designed to deliver Basic to Advance modern web developments which are mainly associated with computer networks, Software UX, and web application development technologies (both the front-end and back-end) for the participants who aim to be Software Engineers. The knowledge gained in this module is helpful for participants to study key areas of web application development in depth.

LEARNING OUTCOMES

At the end of this module, students should be able to:

1. Explain different types of networks and how internet works (HTML & CSS)
2. Describe the usage of web components and how to achieve better user experience
3. Explain software source control and clean code concepts
4. Apply Client-Side Programming (JS & jQuery)
5. Apply Software Architecture Techniques (Java or C#.net with MVC)
6. Apply in Front-end Development Techniques. (Bootstrap & Angular or React)
7. Explain how SEO works (Google Analytics)
8. Explain web hosting, domains and how to work with cPanel

LECTURE SESSIONS

No	Session	Teaching Hours	Learning Outcomes	Method
12.1	Introduction to Internet Technology	6	1	Lecture|Lab
12.2	User Experience	3	2	Lecture|Tute
12.3	Software source control	3	3	Lecture|Lab
12.4	Client-Side Programming	12	4	Lecture|Lab
12.5	Software Architecture	15	5	Lecture|Lab
12.6	Front-end framework / Web Framework	12	6	Lecture|Lab
12.7	Search Engine Optimization	6	7	Lecture|Lab
12.8	Web Hosting	3	8	Lecture|Tute
Total		**60**		

MODULE EVALUATION CRITERIA

Method	Percentage	Learning Outcomes
Course Work -1 [Web-App] [G]	30%	1,2,3,4,5,6,7,8
Final Exam [3 Hours]	70%	1,2,3,4,5,6,7,8

RECOMMENDED REFERENCES

Book	Author(s)	ISBN	Publisher
HTML and CSS Web Standards Solutions	Nicklas Persson Christopher Murphy	978-1-4302-1607-0	Apress
Pro ASP.NET MVC 5	Adam Freeman	978-1-4302-6530-6	Apress
Pro Angular 6	Adam Freeman	978-1-4842-3649-9	Apress
Pro React 16	Adam Freeman	978-1-4842-4451-7	Apress
Introduction to Search Engine Optimization	Todd Kelsey	978-1-4842-2851-7	Apress
Introducing Bootstrap 4	Jörg Krause	978-1-4842-6203-0	Apress
Discussions in User Experience	Dave Lull	978-1-4842-3267-5	Apress

EFFECTIVE COMMUNICATION SKILLS-1

INTRODUCTION

Communication skills are very important for writing documents, facing interviews and conducting presentations successfully. This module is designed to provide knowledge on business communication skills required for industrial environment.

LEARNING OUTCOMES

At the end of this module, students should be able to:

1. Produce/write standard documents
2. Carry on conversations in English
3. Use grammar appropriately

LECTURE SESSIONS

No	Session	Teaching Hours	Learning Outcomes	Method
13.1	Grammar	15	1,3	Lecture|Activities
13.2	Writing	21	1,2,3	Lecture|Activities
13.3	Speaking	16	2,3	Lecture|Presentations
13.4	Reading	06	2,3	Lecture|Presentations
13.5	Listening	03	2,3	Lecture|Activities
	Total	**61**		

MODULE EVALUATION CRITERIA

Method	Percentage	Learning outcomes
Course Work-1[Listening]	20%	2,3
Course Work-2[Speaking]	20%	2,3
Attendance	10%	-
Written Exam [3 Hours]	50%	1,3

RECOMMENDED REFERENCES

Book	Author(s)	ISBN	Publisher
Technical communication: Principles and practice.	Raman, M., & Sharma, S. (2016)	-	Oxford University Press. New Delhi
The ultimate guide to IELTS speaking: The only book with IELTS speaking band descriptors	Thakkar, P. (2007)	-	Parthesh Thakkar.

BUSINESS SOLUTION DEVELOPMENT PROJECT

INTRODUCTION

The business solution development project is one of the core modules in Diploma in Software Engineering program. The project is providing an opportunity for participants to apply concepts, methods and skills acquired during the year to develop business solution to solve business problem.
The project is carried out in groups. Students are not allowed to carry out the project individually. The group size should be **2-4**.

LEARNING OUTCOMES

At the end of this module, students should be able to:

1. Explain the scope of a project
2. Apply methods and collect data of a system in a company
3. Analyze data of an existing system and proposed system in a company
4. Design computerized systems for companies
5. Develop computerized systems for companies
6. Create a Report describing system

CONTENTS

No	Contents	Learning Outcomes
13.1	Project Scope	1
13.2	Data Collection	2
13.3	System Analysis	3
13.4	Design a computerized system	4
13.5	Develop a computerized system	5
13.6	Report Writing	6

MODULE EVALUATION CRITERIA

Method	Percentage	Learning outcomes
VIVA	100%	1,2,3,4,5,6

PROJECT SELECTION AND REGISTRATION

Students are required to design a computerized system for a selected company. When students select a project, scope should be aligned with the group size. The students are allowed to carry out the project in two ways.

- Engineer software for a manual system.
- Engineer existing software to improve an existing computerized system.

After selecting a company and a project title the students are required to meet the course director for registration of the project.

PROJECT SUCCESS PROCEDURE

- Study previous projects in the library
- Periodic discussions with the supervisor
- Walk through the lecture notes before applying theories.
- Effective discussions among project members.
- Active participation in the project

PROJECT DEADLINES AND VIVA

The students must complete the project within the deadline defined by the course director. If a student fails to complete the project within the given deadline, Distinction passes will not be awarded. The following tasks should be carried out according to the defined deadlines.

- Group Registration
- Submission of Project Proposal
- Analysis (DFD/UML/ER)
- Design (File Design /screens/Reports)
- Development of the software using a suitable development language
- Confirmation of the report and the software by the supervisor
- VIVA
- Submission of project Report

INDUSTRIAL SESSIONS

INTRODUCTION

Industrial sessions will be delivered during the program by industry experts to provide the exposure of IT industry and standards used in the industry. This benefits students to understand and align their developments focusing on industry needs. These sessions are organized inside relevant modules.

SESSIONS

Enterprise Application Development	6 Hours
Web Application Development	6 Hours
Database Development	6 Hours
Cloud Application Development	6 Hours
Report Writing and Presentation	6 Hours

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Appendix-1

Examination Rules

Eligibility: - All candidates who possess a minimum of 70% attendance at each modular Lectures/ Discussions of the course are eligible to sit for Examinations.

Those who have below 70% of attendance should follow the balance lecture sessions after making the required payment. They should sit for the examination with the next batch. A candidate who is absent for lecture sessions due to serious injury shall submit a medical certificate obtained from a Medical Officer of a Government Hospital or a recognized Hospital. If they have below 70% attendance should follow balance lectures. They are eligible for marks that they score at the examination.

Instructions to candidates

1. Candidate shall be in attendance outside the examination hall at least 15 minutes before the commencement of examination. He/She shall not enter the Hall until they are requested by the supervisor.

2. On admission to the Hall a candidate shall occupy the seat allocated to him/her and shall not change it except on specific instructions of the supervisor.

3. No candidate shall be admitted to the examination hall for any reason after the expiry of half an hour from the commencement of the examination. A candidate is not allowed to leave the hall until an hour has lasted after the commencement of the examination. A candidate is not allowed to leave during the last 15 minutes of the examination.

4. A candidate should have his/her student Identity Card with him/her in the examination hall on every occasion he/she presents himself for an Examination Paper. The student shall sign a declaration if he fails to produce the Identity Card and produce the Identity card next day. If he/she lost the Identity Card he should obtain a duplicate and produce to the Examination Supervisor. Registrar can make a decision to accept a valid identity card in an emergency situation.

5. No candidate shall keep any notes in his/her clothes or on Identity Card or keep any signs, formulas, or material with him. All notes, books, Parcels, hand bags etc should be kept away and placed at a table indicated by the Supervisor or Invigilator.

6. If the Supervisor so requires each candidate shall declare everything, he/she has in his possession.

7. No candidate shall copy or attempt to copy from any book or paper or notes or similar material or from the scripts of other candidates. No candidate is allowed to talk discuss or ask for any help, answers or any other assistance from the candidate who is seated next to him during the examination time. Candidates should not write any notes, formulas, diagrams and scratches on his body or any part of the body.

8. Nor shall any candidate either help or obtain any assistance from another candidate or person whom so ever. Nor shall any candidate conduct himself so negligently that an opportunity is given to any other candidate to read anything written by him or to watch any papers or any practical examination conducted by him/her. Nor shall any candidate use any other unfair means or obtain or render improper assistance at the examination.

9. Candidate shall bring their own pens, ink, pencils mathematical instrument, erasers or any other equipment which the candidates have been requested to bring to the examination hall.

10. Stationery required for the examination (Writing paper, drawing paper, graph paper etc) will be provided by the institute. Answer sheets or paper provided to a candidate shall not be torn or crumpled or folded or damaged or mutilated. No other papers shall be used by the candidates during the examination. The unused papers or graph tables provided should be used with care and left behind on the desk and not to be taken away from the examination hall.

11. Every candidate shall enter his/her Index Number on the answer book and on every continuation paper. He/She should write on both side of the paper of the booklet and supplementary supplied. He/she shall also enter all necessary particulars as indicated in the cover of the answer book. A candidate who inserts an Index Number other than his own number is liable to be considered as cheating. A script that bears no Index Number or that bears an Index Number which cannot be identified is liable to be rejected. No candidate is permitted to write his/her name or any other identifies and marks on the answer script and no remarks of explanations of any kind (e.g to late arrival or illness) should be addresses to the examiners.

12. All calculations and rough work shall be done on the papers provided and shall be attached to the answer scripts. Such work should not be done on admission card, or on any other paper. Any candidate

who disregards these instructions will be considered as having written notes or outline of answers with the intention of copying.

13. Any answer or part of any answer which is not to be considered for the assessment shall be neatly crossed out. If the same question has been attempted in more than one place it shall be crossed out carefully.

14. All candidates are under the authority of the supervisor during the examination and candidates should assist him/her by obeying and carrying out his/her instruction and assist invigilators before and after the examination.

15. Each candidate shall conduct and behave himself/herself in the Examination Hall and should not cause any disturbance to or inconvenience to the Examination staff or any other candidates. In entering and leaving the hall he/she shall conduct himself/herself as quietly as possible. A candidate is liable to be excluded from the Examination Hall for disorderly conduct.

16. Candidates shall stop work promptly when announced or ordered by the Supervisor or Invigilator.

17. Absolute silence shall be maintained in the Examination Hall. A candidate is not permitted for any reason to communicate or have any dealings with other students other than the Supervisor or Invigilator.
    In case of urgency the candidate can communicate with the Supervisor or Invigilator. The attention can be drawn by raising his/her hand from where he/she is seated.

18. During the course of answering a paper no candidate is permitted to leave the Examination Hall temporarily. In case of an emergency the Supervisor/Invigilator will grant permission to do so but the candidate will be under his supervision.

19. No person shall impersonate a candidate whether in the Examination Hall or before or after the Examination. Nor shall any candidate be allowed to be impersonated by another person.

20. No candidate shall obtain or attempt to obtain prior knowledge of questions or any components of questions presenting in the question paper.

21. Candidates are not allowed to use any mobile communication devices during the examination. If such items are found in their possession during the examination, the devices will be confiscated and returned to them only after the exam offence board makes the decision to return. Students are not allowed to use mobile phones instead on calculator during the examination.

22. Serious Action/Disciplinary Action will be taken if any dishonest assistance is given to a candidate by any person.

23. If any circumstances aroused or under emergency situation in which opinion of the Supervisor renders the cancellation or postponement of the examination is necessary he/she shall stop the examination, collect the scripts written by the student and report matter as soon as possible to Director General (Higher Authority)

24. The supervisor/Invigilator is empowered to obtain a statement in writing regarding any matter, examination offence which has been aroused during the course of the examination conducted and such statement shall be signed by the candidate. No candidate shall refuse to make statement or to sign it.

25. Candidates shall not contact any person other than the Director General or Course Director or Examination Registrar regarding their any matter concerning examinations or Examination Matters.

26. Every candidate must hand over their answer scripts personally to the Supervisor/Invigilator or shall remain in the seat until the answer sheets are collected. Candidates shall not handover their answer scripts to the attendant or any minor employer of the Institute.

27. Face of the candidate should be clearly visible at the examination hall for the identification purposes.

28. The candidate will be required to sign his/her usual signature in the admission card in the presence of the invigilator. Admission card sent to the candidates should be duly perfected and handed over to the invigilator when required to do so.

29. Any candidate who has registered for an examination can withdraw from the examination within specific period of time by submitting medical certificate from registered Government or recognized private hospital at the earliest possible time.

30. If a candidate who was registered for a program could not sit an examination due to ill health, he shall notify it to the Examination Registrar or Course Director immediately. This should be confirmed with supporting documents within seven days to Registrar Examinations or to Course Director.
31. A student who was absent himself/herself for an examination is not eligible for classes or merits unless the results board committee approval is granted.
32. No student shall sit for an examination without any permission or approval obtained from the Course Director or Examination Registrar or Director General of the National Institute of Business Management.

33. Following valid reasons for absence are accepted with documentary evidence for Diploma, Advanced Diploma and Higher Diploma students.

    I. Students serious ill health
    (Accident, Surgery, Confinement etc.)

    II. Death of a member of his immediate Family
    III. Any other grave cause acceptable by the Results Board
    IV. A candidate who is absent for an examination due to medical reason shall submit a medical certificate obtained from a Government Hospital or a recognized Hospital. In the case of hospitalization student shall submit documentary evidence in proof of inability to attend examination within 07 days (Diagnosis card, prescription on discharge or any other proof of hospitalization) after conducting Exam. This Medical Certificate should be endorsed by the Medical Officer of NIBM and it should be presented to the Results Board by the Examination Division.
    V. Absence of a candidate from an examination due to a death of a parent or a sister, brother, spouse or child will be excused if approval is granted by the Results Board on producing appropriate documentary evidence.
    VI. In Case of a Marriage - Documentary evidence should be produced as a proof.

Examination Offences and Punishments

Offences

i. Possession of unauthorized documents (notes, answers) at the time of Examination
ii. Copying during the said examination
iii. Cheating
iv. Removal of stationery from the Examination Hall
v. Disorderly Conduct
vi. Impersonation
vii. Improper Knowledge
viii. Aiding & Abetting
ix. Possession of Electronic Mobile Devices

Punishments

I. Any candidate who violates Examination Rules **5** shall be deemed guilty of the offence of **"Possession of unauthorized documents"** and shall be liable to cancellation of his candidature from the examination and any other punishment decided by the Examination Appeal Board.

II. Any candidate who violates Examination Rules No**7** shall be deemed guilty of the offences of **"Copying"** and shall be liable to cancellation of his candidature from the Examination and be prohibited from sitting examinations of the Institute for a period one year or less than 3 yrs.

III. Any candidate who violates Rule **8** shall be deemed guilty of the offence of having **"Cheated at the Examination"** and shall be liable to the cancellation of his candidature from the examination and should be prohibited from sitting examinations of the Institute for a period of one year to 3 yrs.

IV. Any candidate who is detected of **"Removing stationery provided for the examination"** shall be deemed guilty of an examination offence and shall be liable for a punishment including a cancellation of the examination results and prohibiting from sitting examination of the Institute for one year.

V. Any candidate who violates **14,15,16,17,18** shall be deemed guilty of the **"offence of disorderly conduct"** and shall be liable for punishment. A punishment will be the cancellation of the candidature of examination for a period of one year.

VI. Any candidate who violates Rule **19** shall be guilty of the **"offence of impersonation"** and shall be liable to cancellation and prohibition of sitting examinations of the Institute for 3 years. This is a serious offence and if both candidates are students of the NIBM both should be punished under Examination Rules and under Public Law if necessary.

VII. Any candidate who had violated Rule **20** shall be guilty of an examination offence and shall be liable to be punished by cancellation of the candidature of the examination and further punishments which can be decided by Examination Appeal Board.

VIII. Any candidate who violated Rule 21 shall be guilty of keeping mobile communication devices at the examination hall and shall be liable to be punished by cancellation of the candidature of the examination and further punishments which can be decided by Examination Offence Board.

IX. Any candidate found **"aiding & abetting"** in the commission of any of the above examination offence shall be deemed to have committed that offence and shall be liable to the punishments decided by the Examination Appeal Board.

Examination Offences committed by those other than Candidates

The following actions shall be considered as offences

Dishonest delivery or communication of information

Delivery by any person, fraudulently or dishonestly a secret document or a part thereof or communication of any information relating to the contents of a secret or confidential document or a part of that document to any other person, who is not a person authorized to deliver such document or to communicate such information. Fraudulent or dishonest delivery or transmission of any answer scripts, mark sheet, mark book or any other confidential document relating to an examination to any person who is not a person to whom he is authorized to deliver or transmit such documents.

Divulging of Marks

Divulging by any examiner or any other person entrusted with filing up of Mark Sheets, Mark Books or any other information relating to the answer scripts or Mark Sheets or Mark Books to any other person who is not a person to whom he is authorized to divulge such information.

Dishonest Marking

The marking of an answer scripts fraudulently or dishonestly or incorrectly.

Dishonest Alteration

Using any erasers, chemicals, delete, interpolation or any other alteration in Mark Book, Mark Sheet or Answer Script fraudulently or dishonestly.

Dishonest Assistance

I. Dishonest or fraudulent assistance given directly or indirectly to a candidate to answer a question paper at the examination by a Supervisor, Invigilator, Hall Attendant or any other employee attached to the Examination Hall.

II. Any dishonest erasing, deleting, alternations, interpolation or any other alternation changes made in the answer scripts by Supervisor, Invigilator or by any other employee of the Examination Hall.

Dishonest disposal and disclosure of documents secretly

Secret disposal or disclosure of any confidential document or disposal of any part of the documents secretly by any person who is entrusted with typing, photocopying (Duplicating) packeting or while making manual records or by computerizing any notes or Question Papers of any examination. Or by any other person who is assisting in the above duties while employed in the Examination Unit of the NIBM.

Theft or dishonest breaking

Theft or dishonest breaking of safe or cupboard containing secret documents (Question Papers) Mark Sheets and Mark Books or certificates.

Dishonest breaking and opening of a sealed packet or a safe or a cupboard containing any secret document related to examinations or taking out any such documents without any authority.

Fraudulent Introduction of Answer Scripts.

The introduction or insertion or exchange of another script in place of the script answered by the candidate of an examination of the NIBM.

Dishonestly by Commission or Omission

The commission or omission of any act relating to the conduct of an examination which is deemed by the Director General to be of a fraudulent or dishonest nature.

Assistance or Canvassing

The abetment or assistance with another person in this Institution or another Institution for one or more of the above offences regarding Examination and Results.

Procedure for Disciplinary Action

Disciplinary action shall be taken against those who have committed one or more of the above listed offences. In the case of employees of this Institute Disciplinary action may be taken by Director General including the action by police wherever necessary.

Procedure Regarding Examination Offences Committed by Candidates

Examination Offence Board

There shall be an Examinations Appeals Board comprising of 3 members appointed annually. This committee will inquire in to and make recommendations regarding punishments into Examination Offences referred to it. These offences shall be referred to the committee by the Registrar/Examinations Division on the recommendation of the Director of that discipline concerned and by the Director General.

Classification of the Offences

i. Possession of unauthorized documents (notes & answers)
ii. Copying during the examination
iii. Cheating
iv. Removal of stationery from Examination Hall
v. Disorderly Conduct
vi. Impersonation
vii. Attempting /obtaining improper assistance
viii. Aiding and Abetting
ix. Possession of Electronic Mobile Devices

Punishments

i. Cancellation of Examination and results of the examination
ii. Prohibiting of sitting the examination for one year(specified Period)
iii. Other punishments (decided by the Examination Offence Board)

Procedure

1. Action by Supervisor
2. Action in cases of disorderly conduct
3. Action by Supervisor
4. Further action
5. Offences detected by others
6. Allegation
7. Final Authority

Appendix-2

Lab Rules and Regulations

- Lab users must sign in and sign out in the lab, when entering and leaving the lab respectively.
- Be operate the equipment with respect and care.
- Be silent in the lab.
- No food or drink is allowed in the lab.
- Turn off mobile phones.
- Keep your bags aside or in the rack, if you need to use your mobile phone, please take it out of the lab.
- Do not misuse the hardware.
- If you encounter a problem with hardware or software, inform the instructors.
- Before leaving the lab, you are requested to arrange your chair, monitor, keyboard and mouse in their correct position.
- Any failure to follow these lab rules may result in the loss of your lab privileges.